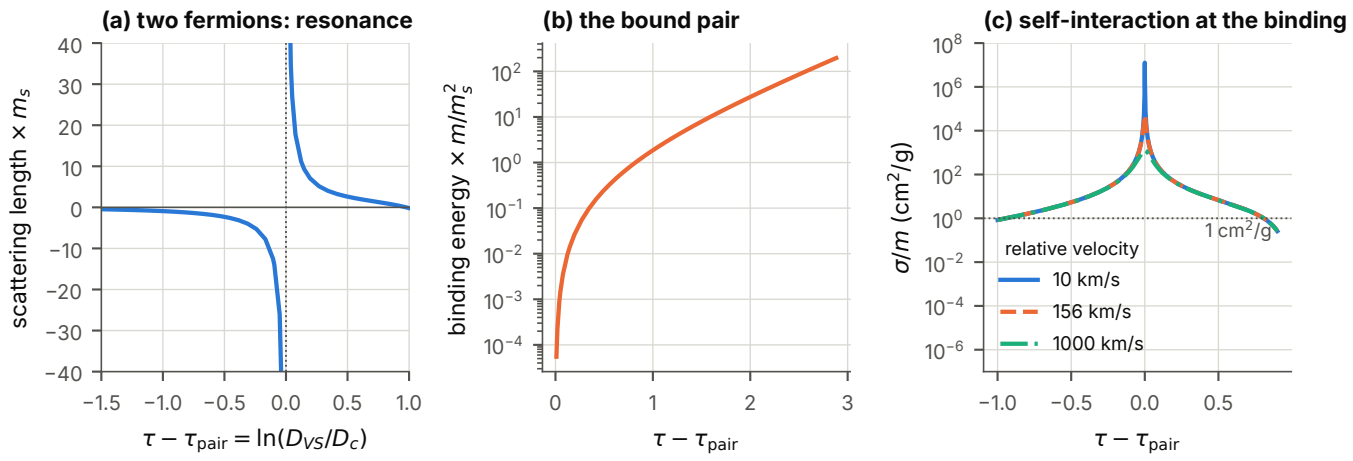
(a) two fermions: resonance
scattering length × m_s
40
30
20
10
0
−10
−20
−30
−40
−1.5
−1.0
−0.5
0.0
0.5
1.0
$\tau - \tau_{\text{pair}} = \ln(D_{VS}/D_c)$
(b) the bound pair
binding energy × m/m_s^2
10^{2}
10^{1}
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
0
1
2
3
$\tau - \tau_{\text{pair}}$
(c) self-interaction at the binding
σ/m (cm^2/g)
10^{8}
10^{6}
10^{4}
10^{2}
10^{0}
10^{-2}
10^{-4}
10^{-6}
1 cm^2/g
relative velocity
10 km/s
156 km/s
1000 km/s
−1.0
−0.5
0.0
0.5
$\tau - \tau_{\text{pair}}$